

Students will be assessed on their ability to:		Suggested experiments
8	draw and interpret free-body force diagrams to represent forces on a particle or on an extended but rigid body, using the concept of <i>centre of gravity</i> of an extended body	Find the <i>centre of gravity</i> of an irregular rod
9	investigate, by collecting primary data, and use $\Sigma F = ma$ in situations where m is constant (Newton's first law of motion ($a = 0$) and second law of motion)	Use an air track to investigate factors affecting acceleration
10	use the expressions for gravitational field strength $g = F/m$ and weight $W = mg$	Measure g using, for example, light gates Estimate, and then measure, the weight of familiar objects
11	identify pairs of forces constituting an interaction between two bodies (Newton's third law of motion)	
12	use the relationship $E_k = \frac{1}{2}mv^2$ for the kinetic energy of a body	
13	use the relationship $\Delta E_{grav} = mg\Delta h$ for the gravitational potential energy transferred near the Earth's surface	
14	investigate and apply the principle of conservation of energy including use of work done, gravitational potential energy and kinetic energy	Use, for example, light gates to investigate the speed of a falling object
15	use the expression for work $\Delta W = F\Delta s$ including calculations when the force is not along the line of motion	
17	investigate and calculate power from the rate at which work is done or energy transferred	Estimate power output of electric motor (see 53)
16	understand some applications of mechanics, for example to safety or to sports	

5.4 Good enough to eat (EAT)

This topic uses a case study of the production of sweets and biscuits:

- measuring and controlling the flow of a viscous liquid
- mechanical testing of products.

There are opportunities for students to develop practical skills and techniques and thus to carry out experimental and investigative activities.

Students will be assessed on their ability to:		Suggested experiments
18	understand and use the terms <i>density</i> , <i>laminar flow</i> , <i>streamline flow</i> , <i>terminal velocity</i> , <i>turbulent flow</i> , <i>upthrust</i> and <i>viscous drag</i> , for example, in transport design or in manufacturing	
19	recall, and use primary or secondary data to show that the rate of flow of a fluid is related to its viscosity	
20	recognise and use the expression for Stokes's Law, $F = 6\pi\eta rv$ and upthrust = weight of fluid displaced	
21	investigate, using primary or secondary data, and recall that the viscosities of most fluids change with temperature. Explain the importance of this for industrial applications	
25	investigate elastic and plastic deformation of a material and distinguish between them	
26	explore and explain what is meant by the terms <i>brittle</i> , <i>ductile</i> , <i>hard</i> , <i>malleable</i> , <i>stiff</i> and <i>tough</i> . Use these terms, give examples of materials exhibiting such properties and explain how these properties are used in a variety of applications, for example, safety clothing, foodstuffs	